ETHNOHACK 2026 · TRACK 1 — TRADITIONAL KNOWLEDGE

# Palimpsest

*reading 2,000 years of herbal medicine against modern evidence*

**Dioscorides** Greek, c. 60 CE **Culpeper** English, 1653 **Modern evidence** 2026

Bridges to Track 2 (drug discovery) and Track 3 (conservation)

**Code & data** github.com/KaraAliOsman/palimpsest

**Explorer** karaaliosman.github.io/palimpsest

Team Palimpsest · EthnoHack 2026

Open the explorer

# Herbals are mined for leads, but resist being read

### THE LANGUAGE

ἰκτερικούς θεραπεύει

"treats the jaundiced": inflected, accented Ancient Greek

### THE PLANTS

ἀψίνθιον = ?

*Artemisia absinthium*? *A. pontica*? *A. maritima*?

### THE DISEASES

dropsy · king's evil · strangury

today: oedema, scrofula, painful urination

### THE EVIDENCE

*"supported", compared with what?*

earlier surveys counted supported claims with no base rate

## Three questions

1. What persisted across 1,600 years of copying?
2. Does persistence carry a modern signal?
3. What has modern science ignored?

# Three layers of text and evidence, 60 CE to 2026




## Dioscorides, *De materia medica*

822

**chapters**

Wellmann's critical edition (First1KGreek TEI). We repaired a nesting bug that hid 199 chapters from parsers.

## Culpeper, *The Complete Herbal*

328

**monographs**

Project Gutenberg #49513. Only the “government and virtues” sections are mined; astrology is ignored.

## Modern literature and data

48.2 M

**Europe PMC records**

plus ClinicalTrials.gov, LOTUS phytochemistry via Wikidata, GBIF, IUCN status and MeSH

# From the original words to one shared key

**DIOSCORIDES · DE MATERIA MEDICA 3.23**

**ἀψίνθιον** … καὶ ἀνορεξίας καὶ **ἰκτερικούς** θεραπεύει

*wormwood … "and treats loss of appetite and the jaundiced"*

**CULPEPER · COMPLETE HERBAL · WORMWOOD**

*"The sun never shone upon a better herb for the **yellow jaundice** than this"*

*1653, "government and virtues"*

**1 NAME CONCORDANCE**

ἀψίνθιον · Wormwood → ***Artemisia absinthium***

confidence high · all names: 99.2% on GBIF; 64% / 72% agree with independent Greek / English checks

**2 INDICATION EXTRACTION**

**ικτερ- · yellow jaundice** → Jaundice

accent-free, offset-preserving Greek stems · 17th-century English disease crosswalk

**SHARED KEY**

*Artemisia absinthium* × Jaundice

MeSH D007565

**claimed in both → persistent**

**85**

MeSH-linked indication concepts

**10,399**

claims, each with a verbatim span

**BLIND REFERENCE EVALUATION · 50 + 49 EVALUABLE RANDOM MONOGRAPHS**

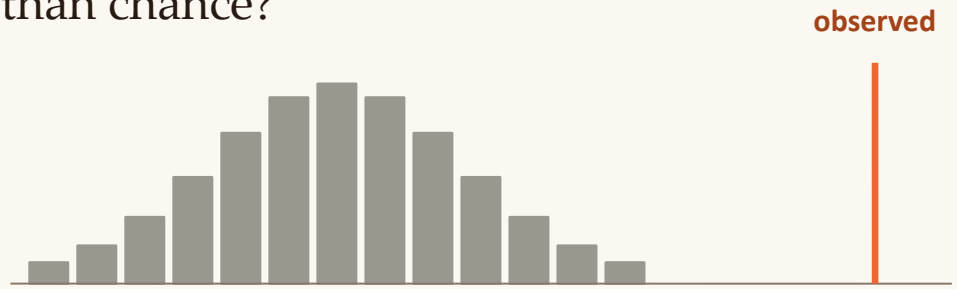
**Greek** precision 0.88 · recall 0.83 · F1 0.86 [0.81–0.89]
**English** precision 0.90 · recall 0.90 · F1 0.90 [0.88–0.92]

*Reference annotation AI-assisted; human double annotation is next.*

# Each question is tested against its own control

Q1 · TRANSMISSION

## Did Culpeper repeat Dioscorides more often than chance?




chance: 10,000 label shuffles

Culpeper's claim sets are shuffled among the shared plants: claims per plant and disease frequencies stay fixed.

Q2 · MODERN SIGNAL

## Is a claimed use studied more than unclaimed uses of the same plant?

*Artemisia absinthium*



claimed — 12 random unclaimed indications, same plant

$$\log \mu = \log E + \beta \cdot group + \gamma_{indication}$$

Logit with indication fixed effects → odds of any study; Poisson GLM with volume offset → depth. Cluster-robust by plant.

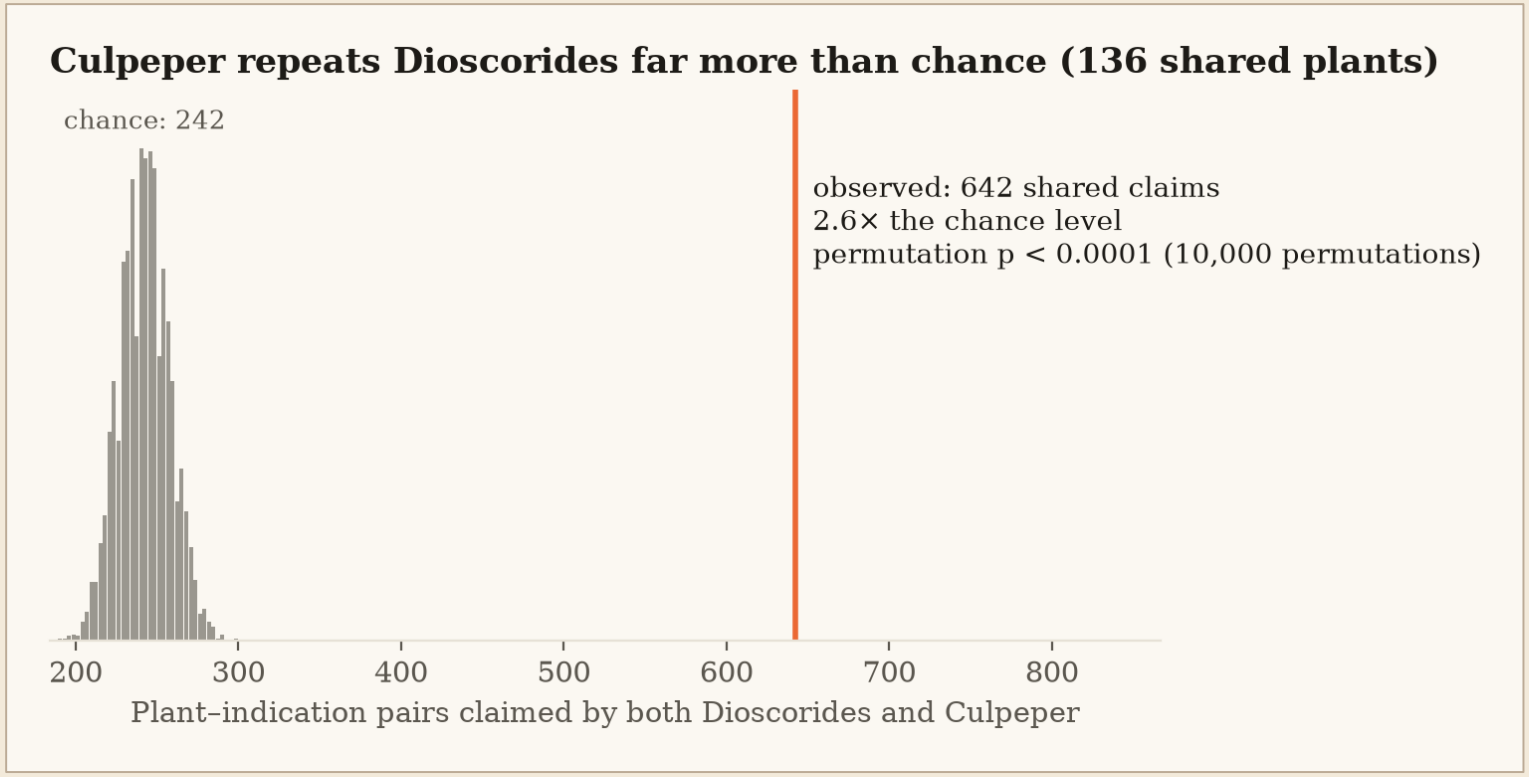
**ROBUSTNESS**

Genus level · high-confidence IDs only · therapeutic categories

**TRANSPARENT LEAD SCORE**

0.35 persistence + 0.25 chemistry + 0.25 gap + 0.15 tractability

# Culpeper repeats Dioscorides 2.6× more often than chance




*136 species in both herbals; 36% of Culpeper's claims for them were already in Dioscorides (13% expected by chance). Persistence of a textual tradition, not independent rediscovery.*

**642**

plant–indication pairs claimed by both authors

**242**

expected if Culpeper's claims were shuffled among the same plants

$p < 0.0001 \cdot z = 26.1$

one-sided, 10,000 permutations

2.1× even when matching plants by entry length; 1.6× even within the same botanical family (both $p < 0.001$)

Genus level 2.39× · high-confidence IDs 2.49×

# Claimed uses are more often studied, not studied more

AT LEAST ONE CO-MENTIONING PAPER

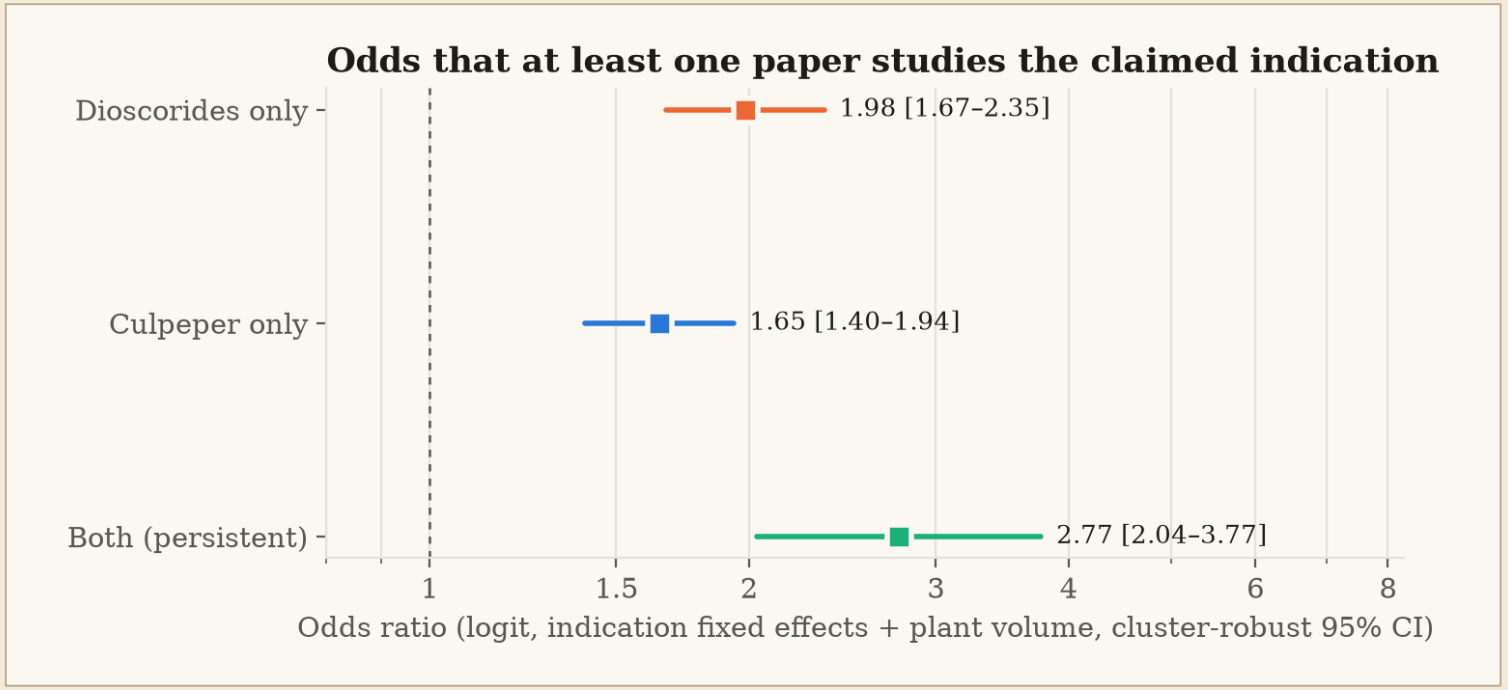
## 45% vs 32%

persistent claims vs matched unclaimed indications of the same plants

| | |
|---|---|
| **Both herbals** | 45% vs 32% · OR 2.77 |
| **Dioscorides only** | 37% vs 25% · OR 1.98 |
| **Culpeper only** | 35% vs 24% · OR 1.65 |

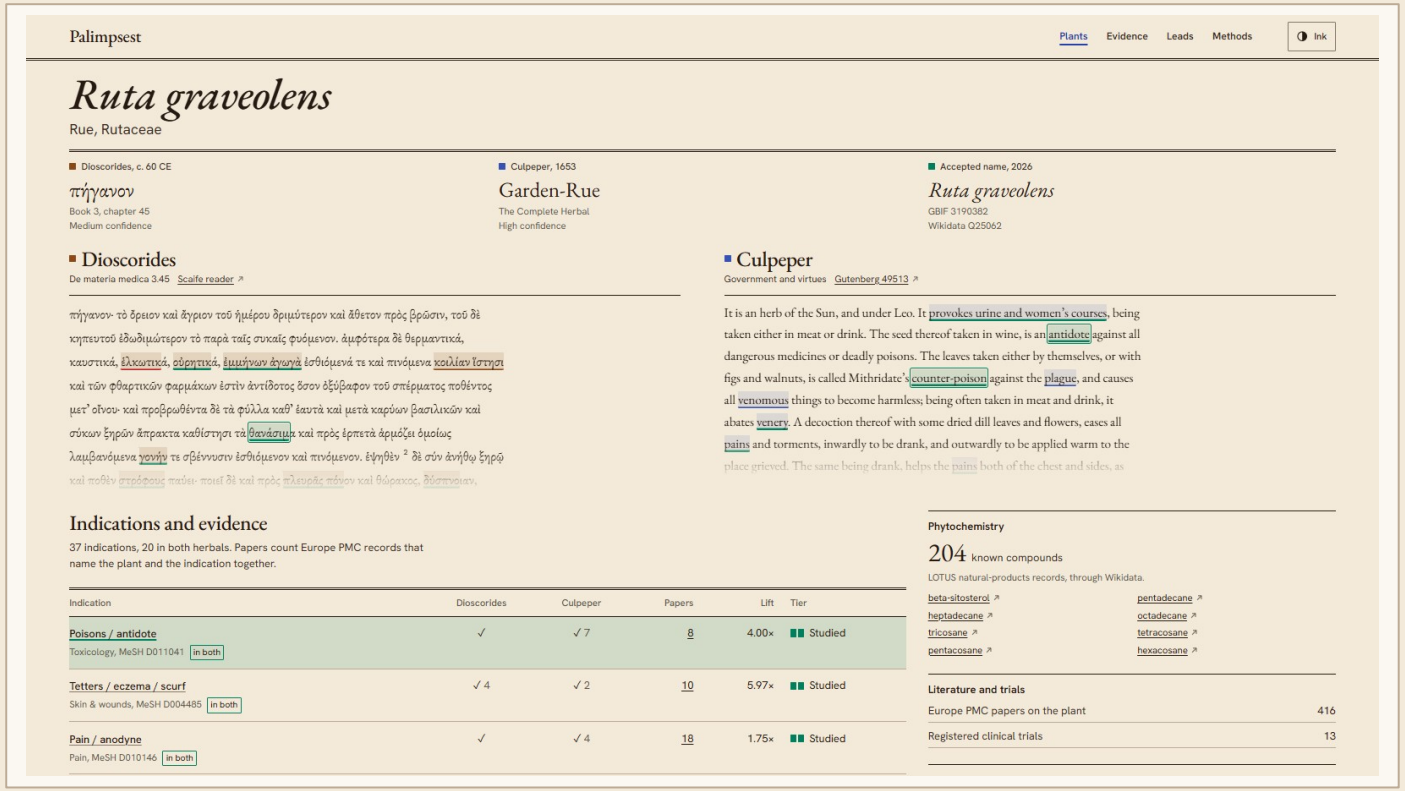
BUT NOT IN DEPTH

Once studied, not studied more: rate ratio 1.04 [0.61–1.76], n.s. That gap is what the leads target.

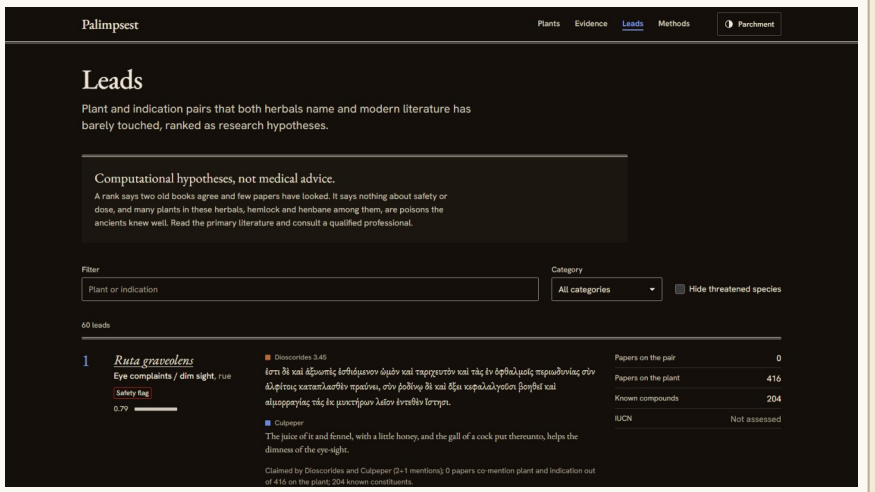



*Logit with indication fixed effects and plant research volume, cluster-robust 95% CI; within-plant Wilcoxon $p = 10^{-15}$. Co-mention measures research attention, not efficacy.*

# Every number links back to the passage it came from




**Plant page** Dioscorides' Greek with each claim highlighted in place, Culpeper alongside, modern evidence per indication




**Leads** ranked, filterable, each score broken into its parts

# 60 persistent but under-studied leads

| # | Plant | Indication | D + C | Papers | Compounds | Safety |
|---|---|---|---|---|---|---|
| 1 | *Ruta graveolens* | Eye complaints / dim sight | 2 + 1 | 0 | 204 | flag |
| 2 | *Vicia faba* | Bleeding / staunching blood | 1 + 1 | 0 | 297 | flag |
| 3 | *Ocimum basilicum* | Venomous bites & stings | 2 + 3 | 0 | 268 | – |
| 4 | *Juglans regia* | Ulcers & sores | 1 + 4 | 0 | 190 | flag |
| 5 | *Cannabis sativa* | Ear-ache / running ears | 1 + 2 | 1 | 528 | flag |

Top 5 of 60 diversified leads from 517 candidates: persistent pairs with ≤ 4 co-mentioning papers. D + C = mentions in Dioscorides + Culpeper; compounds from LOTUS. Flag = the historical text itself warns of harm (shown, never scored).

*Hypotheses for laboratory follow-up, not therapeutic claims.*

**Drug discovery · Track 2**

An explainable shortlist with LOTUS constituents attached

**Conservation · Track 3**

No lead depends on a species assessed as threatened (19 of 60 assessed by IUCN)

**ETHCSTWIN knowledge graph**

20,425 nodes, 51,430 edges: 10,399 claims with quote and offsets, 18,498 compounds

**History of medicine**

Transmission measured passage by passage, from the Greek

# What these results are not, and what comes next

## Responsible research

- Computational hypotheses, never medical advice; several remedies are toxic
- Only published, openly licensed texts; nothing collected from living communities
- AI-assisted curation disclosed; every row has a confidence level and a justification

## Limitations

- Circularity: tradition motivates research, so co-mention ≠ efficacy
- A co-mention may report a null or toxic result
- Ancient plants are identified retrospectively
- Culpeper inherits Dioscorides: persistence, not rediscovery
- Literature searched by accepted binomial only (synonyms implemented, not yet run: rosemary 211 → 2,179 papers)

## Next

- Add Galen, Pliny, Byzantine and Arabic sources to trace the whole chain
- Outcome-aware evidence: classify findings, add ChEMBL targets
- Expert review with ETHCSTWIN, then lab follow-up of top leads

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Code MIT · Greek-derived data CC BY-SA 4.0